Factor-base designs: cost score, third m = 4 cell, small-prime instrument, A11 ablation (run-20261016T0616Z)

Kind: solver-axis stage diagnostic (AGENTS.md §2). Observations only. Nothing here prices a pipeline, reports S or shows a speedup. The protocol is `PREREGISTRATION.md` Amendment 4 (C1–C4), registered before any of its rows. The analysis is `runs/analysis-amendment4.txt` (native `analyze --amendment4`), and every number below is copied from it or from the row files.

Run

- **Commands** (`B=target/release/examples/fb_designs_slope`, `R=research/factor_base_designs_20261015/runs`, run serially):
  ```
  $B replay $R --out C1-rescore --groups p1,F1,F2,F3,F4
  $B run --arms H1,H2 --m 2,3 --file 'C4-{arm}-m{m}' --out-dir $R
  $B run --arms A0,A11 --m 4 --t-min 4 --t-max 4 --budget 3600 --grace 300 --file 'C2-{arm}-m4-t4' --out-dir $R
  $B replay $R --out C3-trace-p3 --groups p1,F2,F3,F4,C2,C4 --only-prime p1 --prime p3
  ```
  The C3 extension ran only if every compared system matched. It did not run (`runs/C3-extension-not-run.txt`).
- **Binary:** built once from `025746fd5`, the Amendment 4 harness commit. Later recorded commits (`236e3bd58`, `86958f03e`) only added rows, Lean files and README lines, with no change under `examples/` or `src/`. Before the run, the extended harness reproduced the committed regression, `analysis.txt` and `analysis-amendment2.txt` byte for byte. `field_ops` was the same on two runs of the same system.
- **Host:** "vm", Intel Xeon @ 2.80 GHz, 4 cores, 15 GB RAM, a Claude Code cloud container. One system at a time, with no further isolation. **Window:** 2026-10-16, 06:16Z to 09:54Z (UTC).
- **Limits:** 300 s F4 budget and 420 s hard kill (C2: 3600 s and 3900 s), `ulimit -v 13000000`. Seeds as in Amendments 1–3, plus prime index $p_3 = 2$ and arms H1 = 7, H2 = 8.
- **Records:** `runs/C1-rescore.jsonl`, `runs/C4-H{1,2}-m{2,3}.jsonl`, `runs/C2-{A0,A11}-m4-t4.jsonl`, `runs/C3-trace-p3.jsonl`, each with its `.log` `runs/analysis-amendment4.txt`.
- **Lean** (outside the measurement): `quartic_norm_factor` in `lean/FactorBaseDesigns/Encodings.lean` proves $Q^2(X^3 + AX + B) - P^2 = (X - x_R)\cdot G$ with G a monic quartic. That is Amendment 3's correction. `lake build`, the axiom audit (standard axioms only) and `leanchecker --fresh` pass. The logs are in `proof_logs/`.

C1, cost score κ = Δlog₂ field_ops / Δb against m − 1

The replay rebuilt and re-measured all 192 committed systems in completed cells (first run, F1, F2, F3, F4). **192 of 192 match their committed rows on D, `max_cols`, `max_rows` and `steps`.** `field_ops` counts F_p multiplications in F4's row reduction only.

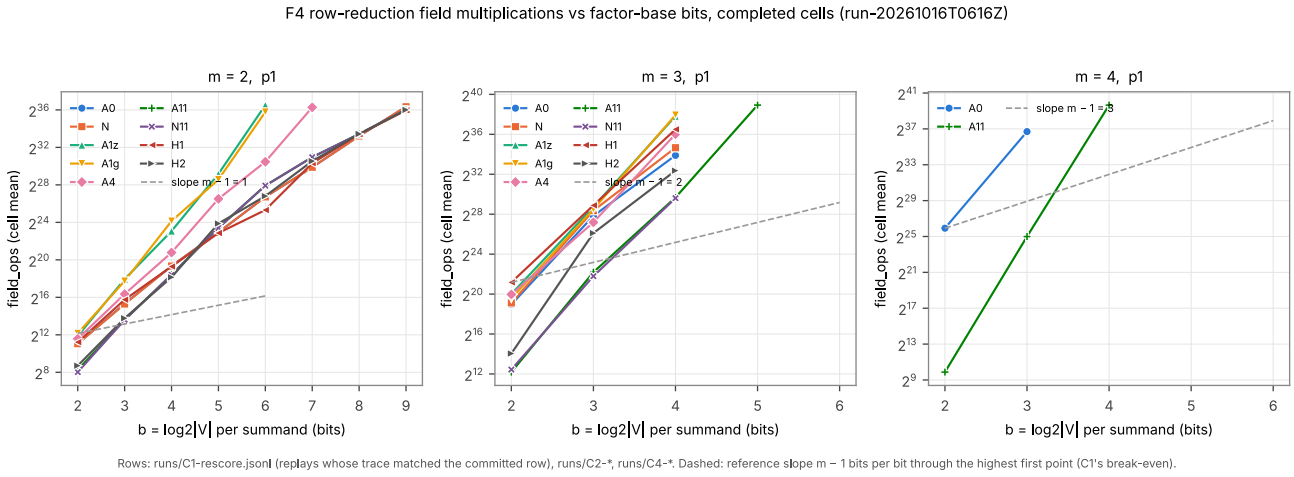


Figure 1: `field_ops` (cell mean of 2 systems, $\log_2$ scale) against b, completed cells on p_1, m = 2, 3, 4. The dashed line has the break-even slope m − 1 bits per bit, drawn through the highest first point. It is a reference, not a fit. Rows from `runs/C1-rescore.jsonl`, `C2-*` and `C4-*`. Run run-20261016T0616Z, `plot.py`.

series	cells	κ	m − 1	σ	C1 verdict
A0 m=2 p_1	8	3.593	1	0.1905	unfavourable
N m=2 p_1	8	3.596	1	0.1905	unfavourable
A1z m=2 p_1	5	6.050	1	0.0000	unfavourable
A1g m=2 p_1	5	5.809	1	0.0000	unfavourable
A4 m=2 p_1	6	4.902	1	0.2286	unfavourable

A11 m=2 p_1	8	3.995	1	0.1786	unfavourable
N11 m=2 p_1	8	4.014	1	0.1786	unfavourable
H1 m=2 p_1	8	3.535	1	0.2619	unfavourable
H2 m=2 p_1	8	3.923	1	0.0000	unfavourable
A0 m=3 p_1	3	7.468	2	0.5000	unfavourable
N m=3 p_1	3	7.772	2	0.5000	unfavourable
A1z m=3 p_1	3	8.905	2	0.0000	unfavourable
A1g m=3 p_1	3	9.189	2	0.0000	unfavourable
A4 m=3 p_1	3	8.028	2	0.5000	unfavourable
A11 m=3 p_1 (with F3, t = 2–5)	4	8.769	2	0.3000	unfavourable
N11 m=3 p_1	3	8.582	2	0.0000	unfavourable
H1 m=3 p_1	3	7.660	2	0.5000	unfavourable
H2 m=3 p_1	3	9.158	2	0.0000	unfavourable
A0 / A11 m=2 p_2	8 / 8	3.595 / 3.995	1	0.1905 / 0.1786	unfavourable
A0 / A11 m=3 p_2	3 / 3	7.468 / 8.742	2	0.5000 / 0.0000	unfavourable
A0 m=4 p_1	2	10.773	3	1.0000	inconclusive (2 cells)
A11 m=4 p_1 (with C2)	3	14.886	3	1.5000	unfavourable

Observations:

- Every decided series is cost-unfavourable. That is 23 series with at least 3 completed cells (22 at m ∈ {2, 3}, plus A11 at m = 4), so the prediction held. κ exceeds m − 1 by 2.5–5 bits per bit at m = 2 and by 5.5–7.2 at m = 3.
- σ and κ disagree on which arms look good. A1z, A1g, N11, H2 and A11 at m = 3 have σ = 0 or near it, yet their κ is at or above A0′s (7.468).
- A11′s κ is above A0′s at m = 2 (3.995 vs 3.593) and at m = 3 (8.769 vs 7.468). Its field_ops are lower at small t. Per bit they grow faster.
- The p_2 κ equals the p_1 κ to within 0.03, consistent with the identical traces of Amendment 2.

C2, the third m = 4 cell (p_1, 3600 s budget)

The lab runner (isolab) has no CLI or credentials in this container, so the cell ran here under the same 13 GB limit. This was registered as a blocker in Amendment 4.

- **A11, t = 4:** both systems completed, each with D = 6 and `max_cols` 88 740, with field_ops $8.630 \cdot 10^{11}$ in 1019.3 s and 1044.1 s. The cell joins F4′s series. A11 at m = 4 now has 3 cells (D = 3, 5, 6): σ = 1.5000, which reads **no middle ground**, and κ = 14.886 against 3, **cost-unfavourable**.
- **A0, t = 4:** both systems aborted on memory under `ulimit -v 13000000`, at 536.9 s and 512.8 s, with no D recorded. A0 at m = 4 stays at 2 cells: **inconclusive**.
- The prediction (A11 completes, A0 does not) held.

C3, p_3 = 40961 as an instrument: not admitted

194 completed p_1 systems with t ≤ 13 (first run, F2, F3, F4, C2, C4) were rebuilt with p_3 seeds. **192 of 194 match** on D, `max_cols`, `max_rows` and `steps`. The 2 mismatches are both A1g, m = 3, t = 3. On p_3 both targets have `max_cols` 5 808 and `max_rows` 5 830. On p_1 the two targets have 5 288 / 5 127 and 5 783 / 5 805. D (8) and `steps` (26) agree.

The rule requires every system to match, so p_3 is **not admitted** and the extension (A0 m = 3, t = 5; A11 m = 3, t = 6; A0 and A11 m = 4, t = 4 on p_3) was not run. The prediction did not hold. Observed alongside: A1g's set $\{h \cdot g^k\}$ has no fixed coset shape (ord(g) is drawn), and its two p_1 targets already differ from each other in width. Every arm built on the coset $g \cdot \mu_2{}^t$ matched. `field_ops` is not comparable between p_3 and p_1: the narrow path counts dense row lengths, while the wide path counts non-zero updates.

C4, the A11 ablation

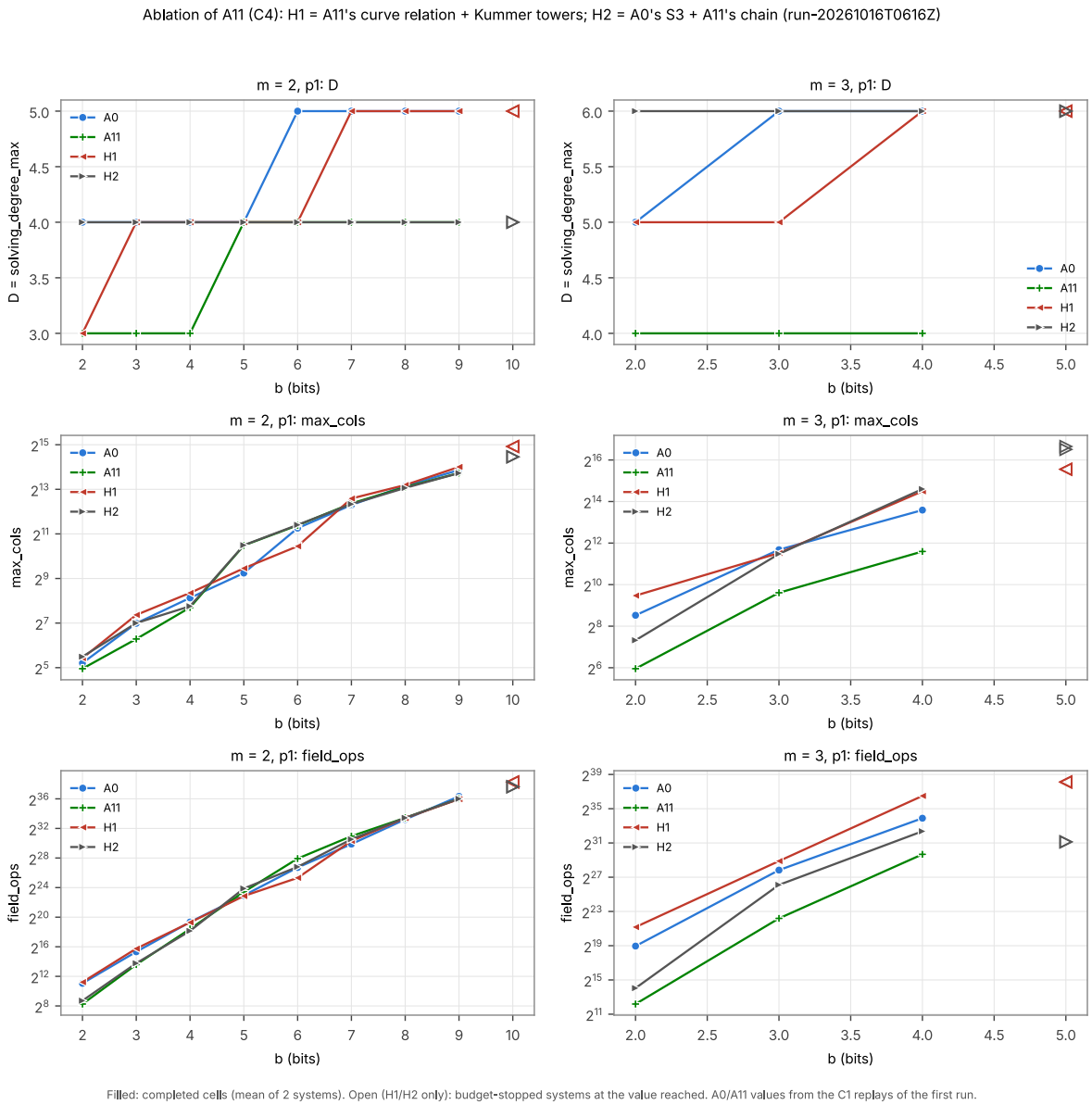


Figure 2: D, `max_cols` and `field_ops` against b at m = 2 (left) and m = 3 (right) for A0, A11 and the hybrids H1 (A11′s coefficient equations with Kummer towers on the roots) and H2 (A0′s S_3 with A11′s squaring chain). Completed cells, mean of 2 systems. Open markers: H1 and H2 systems stopped at the 300 s budget. Run run-20261016T0616Z, `plot.py`.

At m = 3 (D; $\log_2$ field_ops; max_cols per cell):

t	A0	A11	H1	H2
2	5; 18.95; 368	4; 12.19; 62	5; 21.17; 710	6; 14.05; 160
3	6; 27.83; 3 285	4; 22.19; 781	5; 28.89; 2 939	6; 26.10; 2 853
4	6; 33.89; 12 314	4; 29.67; 3 096	6; 36.49; 23 053	6; 32.36; 24 842
5	timeout	5; F3, 3600 s budget	timeout (D ≥ 6)	timeout (D ≥ 6)

Rule outcome: mixed, both on D and on $\log_2$ field_ops (3 shared t). The prediction “the curve relation carries it” did not hold.

- H1 is not cheaper than A0. Its field_ops are higher at every t, and its D is 5, 5, 6.
- H2 has A0′s D (6) or more, and field_ops between A11′s and A0′s at every t. Its width is below A0′s at t = 2 and 3 and above it at t = 4 (24 842 vs 12 314).
- Neither hybrid alone reaches A11′s D or field_ops. In these rows, A11′s lower cost appears only when the function relation and the chain are together. At m = 2 the four arms are within one unit of D of each other and close in field_ops (table in the analysis).

Predictions scorecard (Amendment 4)

prediction	observed	held?
C1: every arm cost-unfavourable at $m \in \{2, 3\}$	22 of 22 decided series unfavourable	yes
C2: A11′s cell completes and A0′s does not	A11 completed (D = 6), A0 aborted on memory	yes
C3: p_3 admitted	192 of 194 match; A1g m = 3, t = 3 differs	no
C4: the curve relation carries it	mixed on D and on field_ops	no

Deviations and limits

- C2 ran on this container, not on a larger machine (blocker above). A0′s t = 4 cell ended on the unchanged memory limit, so A0 at m = 4 still has 2 cells.
- `field_ops` counts row-reduction multiplications only. Symbolic preprocessing, memory traffic and linear algebra over the relations are not counted. κ's break-even (m − 1) assumes relation collection dominates. The verdicts hold at the sizes measured and are not extrapolated.
- Within one cell, the two systems' `field_ops` differ slightly (for example 2 080 268 927 and 2 080 135 234) while D and widths agree. The cell mean is used, as registered.
- The run had no isolation beyond one system at a time. Time-limit stops depend on the host.